

3 Enzymes

AS · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Enzymes lower activation energy by binding substrate at an active site and stabilising a transition state. Initial rate is a fair comparison because substrate depletion and product accumulation are smallest early on. Temperature increases collision rates until heat disrupts protein structure; pH can change ionisation of active-site groups. Competitive inhibitors increase apparent substrate needed; non-competitive inhibitors lower maximum rate.

Worked example: follow the reasoning

A reaction produces 0.30 mmol product in its first 20 s: initial average rate = $0.015 \text{ mmol s}^{-1}$; use a tangent at $t = 0$ for a curved progress graph.

Quick check · multiple choice

A What happens to a competitive inhibitor's effect when substrate increases strongly? [1]

- A It always becomes stronger
- B The enzyme changes into product
- C It may be overcome
- D Temperature becomes irrelevant

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Two enzymes have equal maximum activity, but enzyme X needs higher substrate concentration to reach half-maximal rate. Give one possible explanation and how to test it. [3]

Answers and explanation

Quick check: C. It may be overcome

Question 1 [3 marks]

Competitive inhibition of X is possible [1]; repeat with increasing substrate concentrations [1]; if inhibition can be overcome at high substrate, both approach similar maximum rate [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Biology 9700, 2025–2027. <https://www.cambridgeinternational.org/Images/664560-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.